IFAS: Intelligent Footfall Analysis System for Retail Environments

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***Abstract*—Retail stores record hours of camera footage every day and use almost none of it. This paper presents IFAS, a system that turns that footage into a count of visitors with their gender, their age group, the time they stay and the places they stand. IFAS gives each task the simplest model that does it well. A YOLOv8 detector finds people and an IoU tracker follows them across frames. A ViT-B/16 network classifies gender after transfer learning on the PETA dataset. Four support vector machines classify age from colour and texture histograms, following the original PETA benchmark. Each person is classified on several frames and the results are pooled into one label. In our study the gender model reaches 97.8% validation accuracy on PETA. On the PETA test split the age model reaches a mean accuracy of 0.835 over four age groups and a four-way accuracy of 76.7%. Four out of five of its errors land in a neighbouring age group. We report how sensitive both models are to their main settings and how long each stage takes on a laptop processor. A check on shop footage shows where detection and tracking fail and how far two additions, rejoining lost tracks and a second detector for weak detections, repair it.**

***Index Terms*—footfall analysis, pedestrian attribute recognition, vision transformer, transfer learning, support vector machine, local binary patterns, retail analytics**

I. Introduction

A store that wants to know how many people came in today has two options. It can post someone at the door with a clicker or it can read the number off a sensor above the entrance. Neither says who came in, how long they stayed or where they stood. The cameras that already watch the shop floor record all of that, and almost none of it is used.

Reading the footage by hand does not scale. One person can watch one screen, and a manual count drifts after a few minutes. A store with eight cameras and a twelve hour day produces close to a hundred hours of video a day. The work has to be automated before the footage is worth keeping.

This paper describes IFAS, the Intelligent Footfall Analysis System. IFAS takes a recorded video and returns a record of every person who appears in it. The record holds the person's gender and age group, the time they entered and left the frame and the path they took. A dashboard reads these records and shows them as counts, charts and a short written summary.

The design follows one rule: each task gets the simplest model that does it well. Gender is a binary label with plenty of training data, so a pretrained vision transformer [1] is a good fit. We fine-tune ViT-B/16 in two stages. The first uses the public PETA dataset [2] and the second adapts the model to footage from the target store. Age is harder to read from a pedestrian crop, and PETA has few examples in some age groups. For age we follow the original PETA benchmark [2] and train one support vector machine per age group on colour and texture histograms. This keeps the age results comparable with the published baseline. Detection needs speed more than depth, so a small YOLOv8 model [3] does that job.

The contributions of this work are as follows.

- A pipeline that joins detection, tracking and two attribute classifiers and pools several predictions per person into one label.
- A sensitivity study of four training settings for the gender model and five feature settings for the age model.
- A full evaluation of the age model on the PETA test split, with an analysis of where its errors fall.
- Timings of every stage on a laptop processor, and the per-person cost bound that follows from them.
- A tracker that rejoins people the detector loses, and a second detector that is consulted only about weak detections, with a field check that shows what each one fixes.

Section II reviews related work. Section III describes the system. Section IV reports the experiments. Section V discusses the results and their limits, and Section VI concludes.

II. Related Work

A. Pedestrian Attribute Recognition

Deng et al. [2] introduced the PEdesTrian Attribute (PETA) dataset. It joins ten smaller surveillance datasets into 19,000 images labelled with 61 binary attributes and four multi-class attributes. Gender and age group are among them. Their baseline describes each image with colour and texture features and trains one SVM per attribute. Each attribute is scored by mean accuracy, the average of the accuracy on positive examples and the accuracy on negative examples. A classifier that always predicts the majority class gains nothing under this measure. Later work on pedestrian attributes reports the same measure, and so do we.

B. Vision Transformers

The transformer [4] was designed for text. Dosovitskiy et al. [1] applied it to images by cutting each image into patches

of 16×16 pixels and treating the patches as a sequence. Their Vision Transformer (ViT) matches or beats convolutional networks when it is pretrained on a large dataset such as ImageNet-21k. ViT-B/16 has since become a common starting point for transfer learning. CLIP [5] trains an image encoder together with a text encoder. A binary gender classifier has no use for the text side, so we use a vision-only backbone.

C. Colour and Texture Descriptors

Local binary patterns (LBP) [6] describe texture by comparing each pixel with a ring of its neighbours. The uniform variant is invariant to rotation and to monotonic changes in brightness, and computing it at several radii captures texture at several scales. Colour histograms add the distribution of colour within a region [7]. Support vector machines [8] work well on fixed-length descriptors of this kind. For histograms the intersection kernel is a common choice, and it is the kernel used in the PETA baseline [2].

D. Detection, Tracking and Counting

YOLO [9] predicts every box in an image in one pass, which made real-time detection practical. YOLOv8 [3] is a recent version that includes a small model suited to modest hardware. Counting people by detecting them is easy to audit because every count has a box behind it. Density-map methods estimate the size of a crowd without finding individuals and suit dense crowds better. A store entrance seldom holds more than a few dozen people at once, so detection is enough. Tracking by detection links boxes across frames. SORT [10] showed that a simple association step gives good results when the detector is reliable.

E. Transfer Learning

A pretrained network is usually adapted in one of two ways. Linear probing trains a new output layer on frozen features. Full fine-tuning updates every weight. Kumar et al. [11] showed that fine-tuning from a randomly initialised head can distort the pretrained features and that probing first avoids much of the damage. We adopt this two-phase schedule with AdamW [12] and a cosine learning rate schedule with warmup [13].

III. System Design

A. Overview

Fig. 1 shows the pipeline and Algorithm 1 gives it step by step. A video is read frame by frame. A detector finds people, and a tracker links the detections that belong to the same person. Crops of each tracked person go to the gender classifier and the age classifier. Their outputs are pooled per person. The result is written as one JSON record per video, and the dashboard reads that file.

Algorithm 1 Processing one video

Input: frames $f_0, f_1, \ldots$ at r frames per second; stride s; IoU threshold τ; vote gap g; vote cap m
Output: one record per person

$A \leftarrow \emptyset$ ▷ open tracks
for each frame f_k with $k \bmod s = 0$ **do**
  $B \leftarrow$ person boxes in f_k with confidence ≥ 0.10
  $P \leftarrow \{(t, b) : t \in A,\ b \in B,\ \mathrm{IoU}(t, b) \geq \tau\}$
  for $(t, b) \in P$ in order of falling IoU **do**
    if t and b are both unmatched **then**
      extend t with b
    end if
  end for
  drop unmatched boxes that lie mostly on a matched person
  rejoin an unmatched box to a recently lost track near it
  start a track for each remaining box, tentative if below 0.4
  ask the second detector about tentative tracks that persist
  close each track unmatched for more than r sampled frames
  for each track t matched in f_k **do**
    if t is confirmed, has fewer than m votes and its last vote is at least g old **then**
      crop t from f_k and classify gender and age
      add both probability vectors to the votes of t
    end if
  end for
end for
discard unconfirmed tracks and tracks seen in fewer than three frames
return label, confidence, dwell time and path of each track

B. Detection and Tracking

We use YOLOv8n restricted to the person class with a confidence threshold of 0.4. Boxes smaller than 8×16 pixels are dropped because they hold too little detail to classify. Detection runs on every second frame, which halves its cost.

Tracking uses greedy matching on intersection over union (IoU). For two boxes a and b,

$$\mathrm{IoU}(a, b) = \frac{|a \cap b|}{|a \cup b|}. \tag{1}$$

Every pair of an open track and a new box with an IoU of at least 0.3 is a candidate. Candidates are taken in order of falling IoU, and a pair is accepted if neither side has been matched yet. A track is closed once it has missed as many sampled frames as the video has frames per second, which is two seconds at the default stride. Tracks seen in fewer than three frames are treated as detector noise and discarded.

Two rules keep one person from being counted several times. The first drops a box when half or more of it lies on a person who is already matched in that frame, which removes

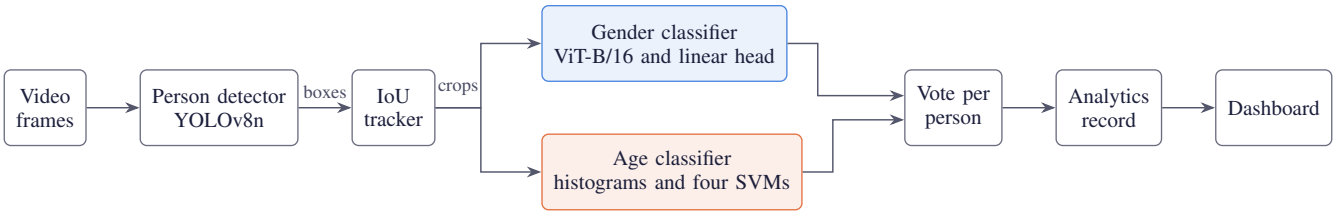


Fig. 1. The IFAS pipeline. People are detected and tracked, each tracked person is classified on several frames by both models, and the pooled labels are written to one record per video.

double detections such as a body and its reflection in glass. The second rejoins lost people. A person cannot vanish in the middle of the frame, so a track that ends there was lost by the detector. A box with a confidence of at least 0.25 that appears within one body height of that spot within 8 s continues the old track. Near the edges of the frame people do leave and others arrive, so there a box rejoins a track only if that person was standing still and the box is in the same place.

The tracker has no appearance model. It is cheap, and Section V-E lists what it cannot do.

C. Weak Detections and the Second Detector

The detector is run with a floor of 0.10, well below the 0.4 needed to start a track. Boxes in between are used in two ways. A weak box that overlaps a person already being followed extends that track, since a weak box on a known person is usually real. A weak box anywhere else opens a tentative track. A tentative track is followed like any other and counts for nothing.

A tentative track becomes a person in one of two ways. A later box of 0.4 or more on the same track confirms it. Failing that, once the track has eight detections a second and larger detector, YOLOv8m, is run on the same frame at input sizes of 640 and 960 pixels. If it finds a person with a confidence of at least 0.25 on that spot, the track is confirmed. The question is asked at most four times per track and at most once a second. A track that is never confirmed is dropped, and its location is remembered so that the same object is not asked about again.

A confirmed track keeps its history, so the dwell time of that person starts at the first weak sighting. If the track began where a known person was lost, the two are merged.

The reason for this arrangement is cost. Section IV-G shows that the larger model finds people the small one misses and is six to eleven times slower. Asking it only about tentative tracks buys most of its benefit for a small part of its cost.

D. Gender Classifier

Each crop is resized to 224×224 pixels and normalised with the ImageNet mean and standard deviation. The backbone is ViT-B/16 pretrained on ImageNet-21k, with 86 million parameters. It splits the crop into 196 patches, adds a class token and passes the sequence through 12 encoder layers. A pooling layer maps the class token to a vector of 768 dimensions. One linear layer with 1,538 parameters maps that vector to two logits, and a softmax turns them into probabilities (Fig. 3a). The loss is cross-entropy.

TABLE I
TRAINING SCHEDULE OF THE GENDER MODEL

Stage	Phase	Trained	Epochs	Peak rate
1, PETA	A, probe	head	4	10^{-3}
	B, fine-tune	whole network	10	10^{-5}
2, store	A, probe	head	2	10^{-4}
	B, fine-tune	whole network	5	10^{-6}

AdamW, batch size 64, weight decay 0.3, warmup ratio 0.2, gradient clipping at 1.0, mixed precision

Training runs in two stages, and each stage has two phases (Table I). Phase A is a linear probe. The backbone is frozen and only the new head is trained, with a learning rate of 10^{-3}. Phase B unfreezes the backbone and trains the whole network with a learning rate of 10^{-5}. Stage 1 runs both phases on PETA. Stage 2 repeats them on labelled crops from the target store with both learning rates divided by ten, so that the model adapts without losing what it learned from PETA.

Each phase has its own schedule. The learning rate rises linearly from zero over the first part of the phase and then falls to zero on a cosine curve (Fig. 2). With T steps in the phase and a warmup ratio w, the rate at step i is

$$\eta_i = \eta_{\max} \cdot \begin{cases} \dfrac{i}{wT} & i < wT, \\ \dfrac{1}{2}\left(1 + \cos \dfrac{\pi\,(i - wT)}{T - wT}\right) & i \geq wT. \end{cases} \tag{2}$$

All runs use AdamW [12] with a batch size of 64, mixed precision and gradient clipping at 1.0. Training images are flipped horizontally with probability 0.5 and receive a small random change in brightness, contrast and saturation. One tenth of the data is held out for validation with a stratified split.

E. Age Classifier

The age model uses no neural network. Each crop is cut into R horizontal regions of equal height. With four regions they correspond roughly to the head, the torso, the upper legs and the lower legs. Two kinds of histogram are computed for each region. The first is colour: one histogram of B bins for each of the red, green, blue, hue and saturation channels. The second is texture: one histogram of uniform LBP codes for each radius in a set $\mathcal{S}$, computed with Q sampling points and therefore $Q + 2$ bins. Every histogram is L1-normalised. The

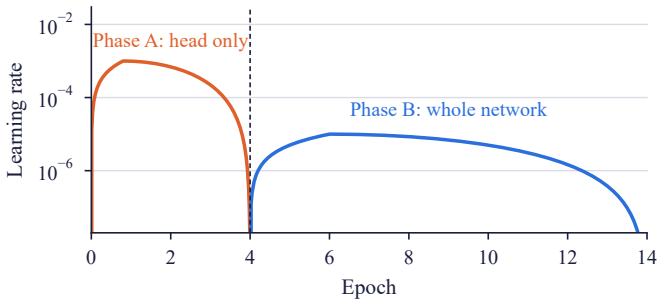


Fig. 2. Learning rate over Stage 1 with the selected settings. Each phase warms up for a fifth of its steps and then decays on a cosine curve. The vertical axis is logarithmic.

histograms of all regions are concatenated into one feature vector (Fig. 3b) of length

$$d = R\left(5B + |\mathcal{S}|\,(Q+2)\right). \tag{3}$$

One SVM is trained for each of four age groups: 16 to 30, 31 to 45, 46 to 60 and above 60. Each is a binary classifier that separates its group from the other three. For every group we search over two kernels. The histogram intersection kernel is

$$K(\mathbf{x},\mathbf{y}) = \sum_{i} \min(x_i, y_i), \tag{4}$$

and the radial basis function (RBF) kernel is

$$K(\mathbf{x},\mathbf{y}) = \exp\!\left(-\gamma\,\|\mathbf{x}-\mathbf{y}\|^2\right). \tag{5}$$

The search covers $C \in \{0.01, 0.1, 1, 10, 100\}$ for both kernels and $\gamma \in \{0.001, 0.01, 0.1, 1\}$ for the RBF kernel, 25 settings in all. The setting with the highest validation mean accuracy is kept for that group. Mean accuracy for one attribute is

$$\mathrm{mA} = \frac{1}{2}\left(\frac{TP}{P} + \frac{TN}{N}\right), \tag{6}$$

where P and N are the numbers of positive and negative examples and TP and TN are the numbers of each that are classified correctly. At test time a crop receives the label of the SVM with the largest decision value.

PETA also labels people under 15. That group has too few examples to train a stable classifier, and early models underfitted it every time. We left it out.

F. Pooling Predictions per Person

A single crop can be blurred or partly hidden. A person is visible for many frames, so IFAS classifies each track several times and combines the answers. A track is classified at most once every 0.4 s and at most 12 times in total. Spacing the samples keeps them from being near copies of one another, and the cap bounds the cost of a person who stands in view for minutes.

Let $\mathbf{p}_1, \ldots, \mathbf{p}_n$ be the probability vectors a model returns for the n crops of one track. Each crop casts a vote for its most probable label. The track receives the label $\hat{c}$ with the most votes, and a tie goes to the label with the higher summed probability. The confidence reported for the person is the share of the summed probability held by that label,

$$\mathrm{conf} = \frac{\sum_{j=1}^{n} p_{j,\hat{c}}}{\sum_{j=1}^{n}\sum_{c} p_{j,c}}. \tag{7}$$

SVM decision values are passed through a softmax first, so that age confidence is on the same scale as gender confidence.

TABLE II
DWELL TIERS

Tier	Time in frame	Reading
Passing	under 3 s	walked through
Glancing	3 to 10 s	slowed or looked
Browsing	10 to 30 s	stopped
Engaged	30 to 60 s	stayed
Dwelling	over 60 s	waited or queued

G. Analytics and Dashboard

From each track IFAS takes the time the person was first seen, the time they were last seen and the dwell time between the two. Dwell time is sorted into the five tiers of Table II. The video is divided into bins of 5 s. For each bin the record holds the number of arrivals, the number of exits and the mean and peak number of people in view. A grid of 16×9 cells over the frame accumulates where people stood, and the frame is split into three vertical zones.

Recordings differ in length, so IFAS compares them by rate. Every recording is split by the clock hours it covers, and each hour h stores the arrivals a_h and the minutes m_h that were watched. The rate for any group H of hours, such as all Saturdays, is

$$\rho_H = 60\,\frac{\sum_{h\in H} a_h}{\sum_{h\in H} m_h} \tag{8}$$

arrivals per observed hour. A three minute clip then cannot outweigh a three hour export. A day needs at least five minutes of footage to count, and a trend is drawn only when three such days exist.

For a coarser summary a recording can also be divided into five equal intervals. Each interval reports the mean, peak and minimum number of people in view and the standard deviation of that number. A low standard deviation marks steady flow and a high one marks a surge. Each interval receives one of six density labels from Empty to Overcrowded. This labelling is at an earlier stage than the rest of the system and is not evaluated here.

A dashboard presents the record at two levels of detail. The simple level states the result in sentences and shows a few headline charts for whoever runs the store. The detailed level adds filters by gender, age, dwell tier and confidence, a table of every tracked person and the confidence distributions of both models.

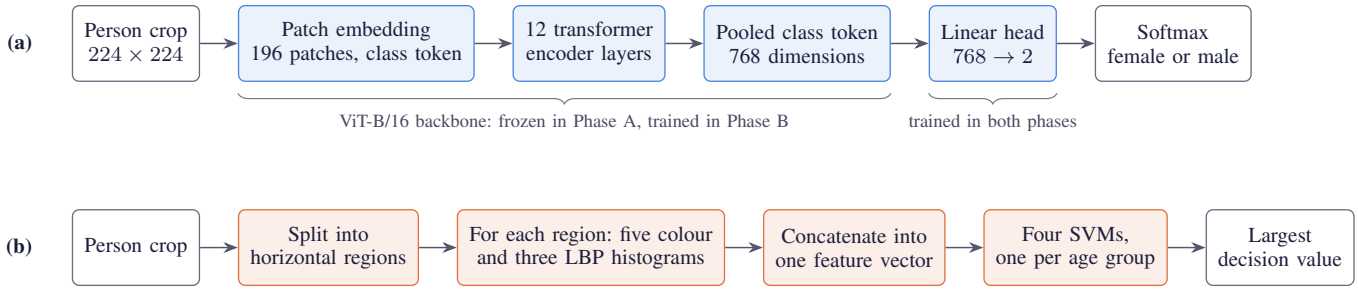


Fig. 3. The two attribute classifiers. (a) The gender network: a pretrained ViT-B/16 with a new linear head. (b) The age model: region histograms scored by one SVM for each age group.

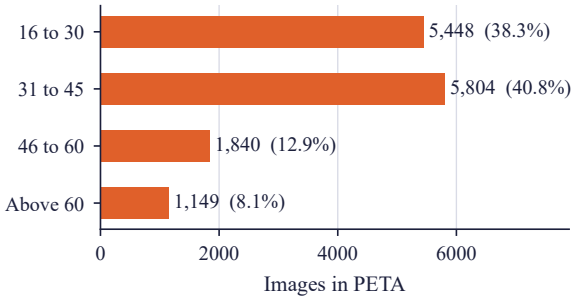


Fig. 4. Images per age group among the 14,241 PETA images used for the age model.

TABLE III
STAGE 1 SETTINGS TESTED FOR THE GENDER MODEL

Setting	Values tested	Selected
Backbone learning rate	10^{-4}, 10^{-5}, 10^{-7}	10^{-5}
Linear probe epochs	0, 1, 2, 3, 4	4
Weight decay	0, 0.001, 0.01, 0.05, 0.1, 0.3, 0.5	0.3
Warmup ratio	0, 0.05, 0.1, 0.2, 0.3, 0.4	0.2

IV. Experiments

A. Data

All experiments use PETA [2]. The gender model uses every image with a gender label, about 19,000 in all, with one tenth held out for validation.

The age model uses the 14,241 images that carry one of the four retained age labels. Fig. 4 shows how they are spread. The two youngest groups hold 79% of the images and the oldest holds 8%. We split the images into training, validation and test sets in the proportions of the PETA protocol, 9,500 to 1,900 to 7,600. This gives 7,120 training, 1,424 validation and 5,697 test images. The split is stratified by age group.

B. Gender Model

We varied four settings of Stage 1 one at a time: the backbone learning rate, the number of linear probe epochs, the weight decay and the warmup ratio. Table III lists the values and Fig. 5 shows the results. Schedules were short, between three and seven epochs in total, and each setting was trained once.

Learning rate. With a backbone learning rate of 10^{-7} the model barely moves. Validation accuracy stays near 0.80, the level the linear probe reaches alone. Rates of 10^{-4} and 10^{-5} both converge, to 0.992 and 0.983 after four epochs of fine-tuning (Fig. 5a). The higher rate scores better here, but the run is short. We chose 10^{-5} because a smaller step is less likely to damage the pretrained features over the longer schedule of the final model. These runs fine-tuned for one epoch longer than the probing and weight decay runs, so their accuracy is not comparable with the panels that follow.

Linear probe. Skipping the probe gives the lowest accuracy, 96.8%. Any amount of probing helps, and four epochs give the highest accuracy at 97.8% (Fig. 5b). The trend is not monotonic, since two epochs score below one. With one run per setting we read that dip as noise. The training loss points the other way. Without a probe the loss over the first 200 steps of fine-tuning falls lower than with one. Lower training loss with lower validation accuracy is what we would expect if the network fits the training set at the expense of its pretrained features [11].

Weight decay. Accuracy is lowest without weight decay at 97.39% and highest at a decay of 0.3 with 97.71%. The gap between training and validation accuracy is also smallest at 0.3: 1.67 points against 2.14 points without decay (Fig. 5c and d). The whole range spans a third of a point. The effect is small, but both measures point the same way.

Warmup. Training without warmup gives the lowest accuracy, 96.86%. A ratio of 0.2 gives the highest at 97.68%, with 0.05 close behind at 97.61% (Fig. 5e). The values in between follow no clear order.

Taken together, the probe moves accuracy by 1.0 point across its range, warmup by 0.8 and weight decay by 0.3.

C. Age Model: Feature Settings

Feature settings were screened on a 30% subsample of the training and validation data to keep the search fast. The settings were tuned in the order listed in Table IV, and each choice was fixed before the next was tuned. Fig. 6 shows the average mA over the four age groups.

The number of body regions matters most. Average mA rises from 0.7155 with two regions to 0.7457 with four and then falls slightly (Fig. 6a). Regions thinner than a quarter of the body hold too few pixels for a stable histogram. Colour

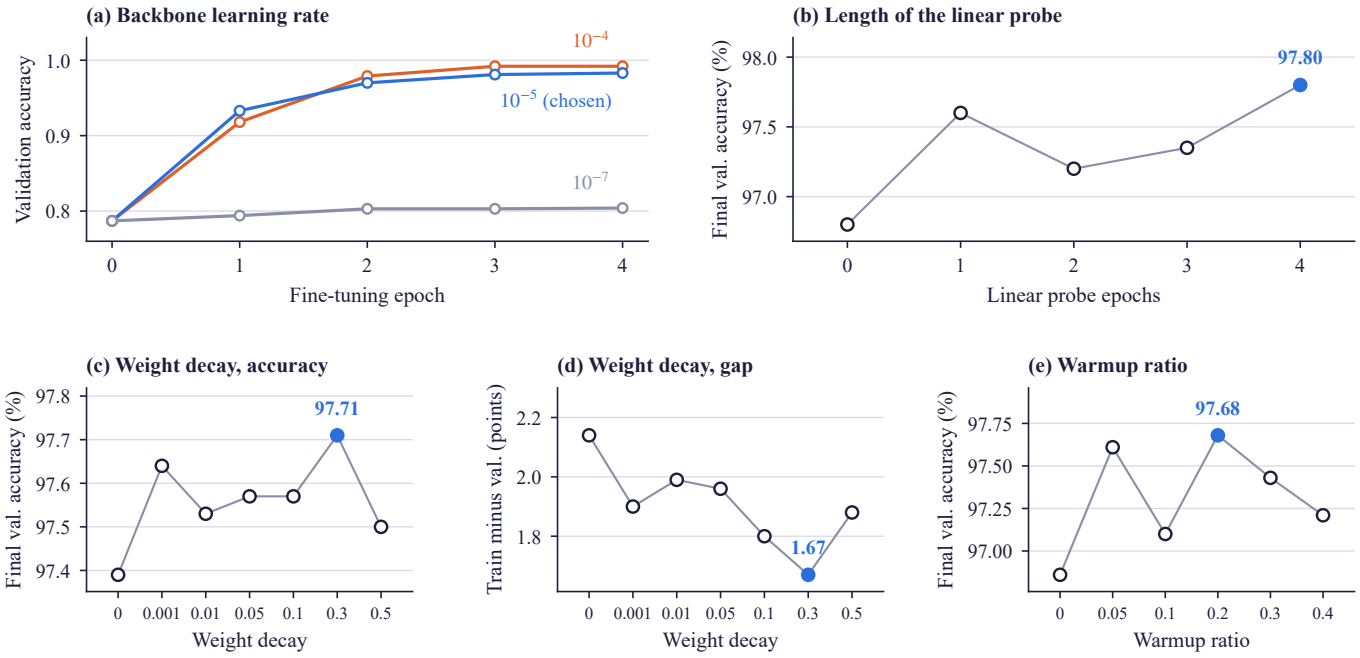


Fig. 5. Sensitivity of the gender model to four training settings on PETA. The selected value is filled in blue. (a) Validation accuracy during fine-tuning for three backbone learning rates. Epoch 0 is the end of the linear probe. (b) to (e) Final validation accuracy, or the gap between training and validation accuracy, with one setting varied and the rest fixed. Each point is a single run.

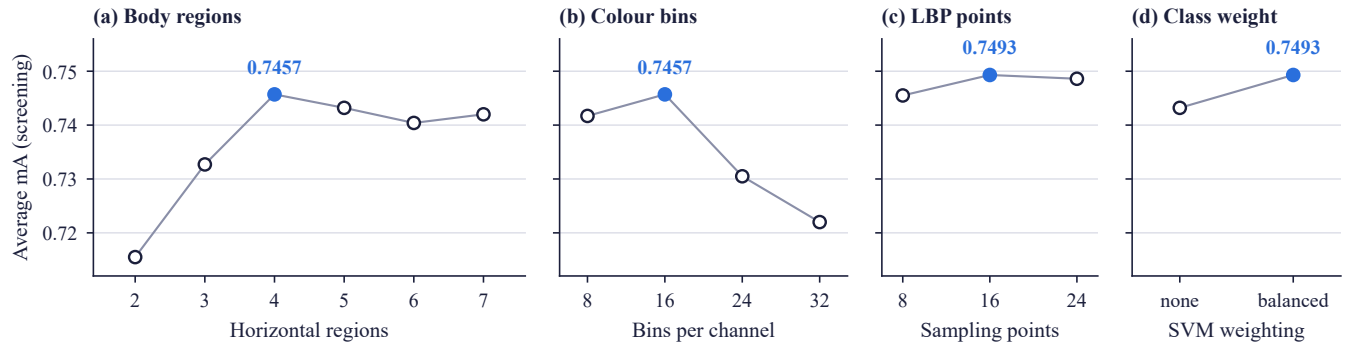


Fig. 6. Sensitivity of the age model to its feature settings, measured as average mA on a 30% subsample of PETA. All four panels share one vertical scale. The selected value is filled in blue.

TABLE IV
FEATURE SETTINGS TESTED FOR THE AGE MODEL

Setting	Values tested	Selected
Body regions R	2, 3, 4, 5, 6, 7	4
Colour bins B	8, 16, 24, 32	16
LBP points Q	8, 16, 24	16
LBP radii $\mathcal{S}$	{1}, {1,2}, {1,2,3}, {1,2,3,4}, {2,3}	{1,2,3}
Class weight	none, balanced	balanced

bins behave the same way. Sixteen bins score 0.7457, and 32 bins drop to 0.7220 as the histograms become sparse (Fig. 6b). The number of LBP sampling points has a smaller effect, with 16 points best at 0.7493 (Fig. 6c). The five combinations of LBP radii we tried scored within a narrow band, and radii of 1, 2 and 3 were marginally best. Balanced class weights raise average mA from 0.7432 to 0.7493 (Fig. 6d), as expected given the imbalance in Fig. 4.

D. Age Model: Final Classifiers

We trained the final SVMs on the full training set in two configurations (Table V). The first is the one released with the pipeline. It uses five regions and eight LBP points, which by (3) gives 550 features per crop. The second uses the settings selected above, four regions and 16 points, for 536 features. The grid search chose the RBF kernel for every group in both. The two configurations are close on the validation split, with a mean mA of 0.821 and 0.823. The selected one is better for the oldest group, 0.920 against 0.911.

TABLE V
FINAL AGE CLASSIFIERS: KERNEL SETTINGS AND VALIDATION MA

	Released, $d = 550$			Selected, $d = 536$		
Age group	C	γ	mA	C	γ	mA
16 to 30	1	1	0.7996	1	1	0.8005
31 to 45	10	1	0.7823	1	1	0.7829
46 to 60	10	0.1	0.7897	10	0.1	0.7866
Above 60	10	0.1	0.9108	10	0.1	0.9204
Mean			0.8206			0.8226

TABLE VI
RELEASED AGE MODEL ON THE PETA TEST SPLIT

Age group	Images	Precision	Recall	F1	mA
16 to 30	2,179	0.730	0.812	0.769	0.8107
31 to 45	2,322	0.794	0.720	0.755	0.7888
46 to 60	736	0.730	0.728	0.729	0.8312
Above 60	460	0.897	0.848	0.872	0.9102
All	5,697	accuracy 0.767, macro F1 0.781			0.8352

E. Age Model: Test Results

The released classifiers were evaluated once on the 5,697 held-out test images. Table VI and Fig. 7 give the results. Mean accuracy is 0.8352 averaged over the four groups, slightly above the validation figure of 0.8206. The largest change is for the 46 to 60 group, which rises from 0.790 to 0.831.

Read as a single four-way classifier, the model labels 76.7% of test images correctly, with a macro-averaged F1 of 0.78. The oldest group is the easiest, with a precision of 0.90 and a recall of 0.85. The 46 to 60 group is the hardest, with an F1 of 0.73.

The confusion matrix in Fig. 7a shows where the errors go. Of the 1,330 misclassified images, 1,057 are placed in a group next to the true one. That is 79.5% of all errors. Counting those as near misses, 95.2% of test images land in the true group or a neighbouring one. The single largest confusion is between the two youngest groups: 23% of people aged 31 to 45 are labelled 16 to 30, and 14% the other way.

F. Runtime

We timed each stage on a laptop with an AMD Ryzen 3 3250U processor, two cores and no graphics card. The input was a recording of 848×478 pixels, and the classifiers were timed on the person crops the detector found in it. Fig. 8 shows the median times. Detection takes 157 ms per frame. The age model takes 55 ms per crop, of which 43 ms is spent computing histograms. The gender network takes 794 ms per crop in a batch of 16 and 900 ms for a single crop, and it dominates the cost. This is why votes are capped. With 12 votes the two classifiers cost under 12 s per person on this machine, no matter how long the person stays in view. Without the cap the cost would grow with dwell time, since at the default stride a person is detected about a dozen times a second. On a processor of this class IFAS is a tool for recorded footage. A live feed would need faster hardware.

TABLE VII
SHARE OF 32 FRAMES IN WHICH EACH PERSON IS DETECTED AT A CONFIDENCE OF 0.25 OR MORE

Detector, input size	ms	Seated	Walking	Left desk	Behind glass
YOLOv8n, 640	183	100%	50%	38%	0%
YOLOv8s, 640	433	100%	62%	6%	0%
YOLOv8s, 960	890	100%	53%	3%	12%
YOLOv8m, 640	1,057	100%	72%	66%	0%
YOLOv8m, 960	2,084	100%	75%	25%	44%
YOLOv8n, 640, at 0.05	183	100%	81%	100%	100%

ms is the time per frame on the laptop of Section IV-F. The last row also produced 50 boxes that lie on no person.

G. Field Check on Shop Footage

PETA measures the classifiers. It says nothing about detection or tracking, so we ran the whole pipeline on a recording from outside the dataset: 15 s of a shop interior at 1280×720 pixels, with glass partitions and stacked boxes. We labelled it by eye. Five people are in view for the whole clip. A man sits in the open. A second man walks across the room and works behind a glass partition. A third person sits at a desk at the left edge of the frame. A fourth sits behind glass at the back. A fifth stands almost fully hidden behind the third.

Detection. Table VII shows how often four of these people are detected in 32 evenly spaced frames. The fifth is never detected as a separate person by any setting. No setting finds everyone. The small model at its normal threshold never finds the person behind glass. The medium model at 960 pixels finds that person in 44% of frames and loses the person at the left desk. At a threshold of 0.05 the small model fires on all four people, but it also fires on a display stand, and with a higher confidence than on the person behind glass. A lower threshold alone cannot tell the two apart.

Tracking. Table VIII compares two versions of the pipeline on this clip. The first uses IoU matching alone with a single threshold of 0.4. The second adds the rules of Sections III-B and III-C. The first version reports five visitors, which happens to equal the true number and is wrong in every detail. The walking man is counted three times, because the detector loses him behind the glass. Two people are missed. The mean dwell time is 4.3 s for people who were present for 15 s, and three of five tracks are labelled as passing through.

The second version reports four visitors, and each is a different real person. The walking man is one track. The person at the left desk is followed for the whole clip, and the person behind glass is confirmed by the second detector. The display stand is opened as a tentative track twice and rejected both times. The second detector was consulted on 11 of the 188 sampled frames. Processing time on the laptop rose from 77 s to 205 s.

This is one clip, chosen because it is hard, and the rules were tuned on it. It shows what kind of error the first version made and what kind of rule removes it. It is not a measurement of accuracy. The classifiers also did less well here than on PETA. Gender was stable for the three people who are clearly visible

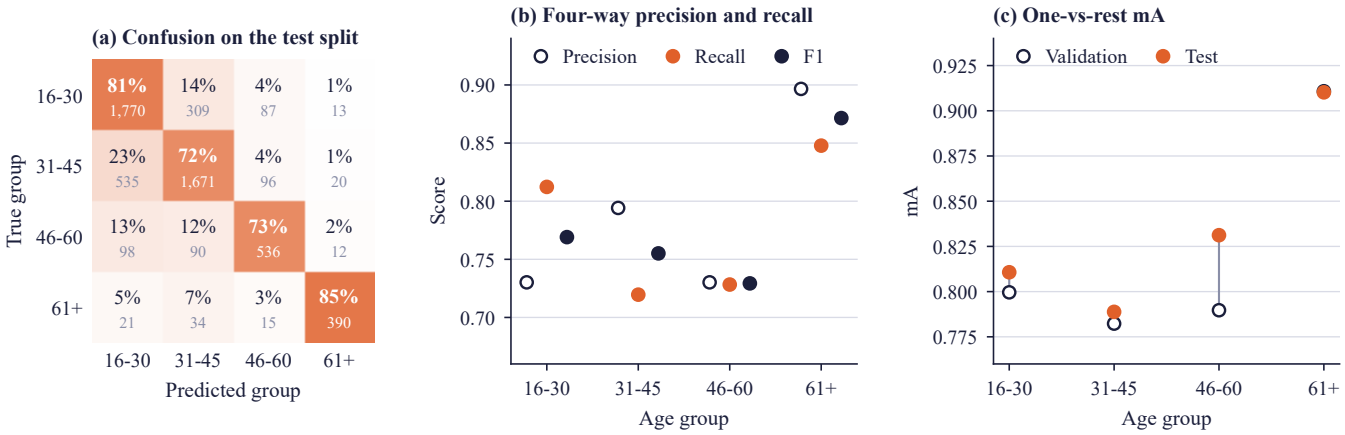


Fig. 7. The released age model on the PETA test split. (a) Confusion matrix. Each row sums to 100%, and the small figure is the number of images. (b) Precision, recall and F1 when the four SVMs act as one four-way classifier. (c) One-vs-rest mean accuracy on the validation and test splits.

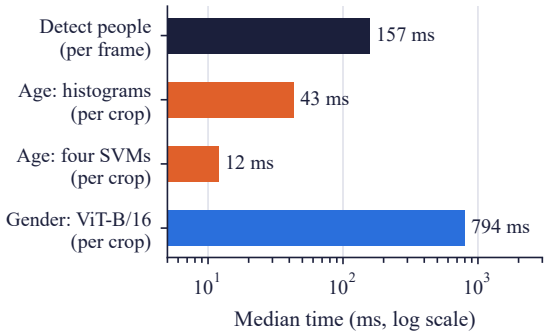


Fig. 8. Median time per stage on a two-core laptop processor. The horizontal axis is logarithmic. Detection is per frame and the classifiers are per person crop.

TABLE VIII
THE PIPELINE ON THE SHOP CLIP, BEFORE AND AFTER THE TRACKING RULES. EACH CELL IS THE DWELL TIME REPORTED FOR THAT PERSON.

Person (present for 15 s)	IoU matching only	With all rules
Seated in the open	15.0 s	15.0 s
Walking, then behind glass	3.4, 1.5 and 1.0 s	14.7 s
At the left desk	0.6 s	15.0 s
Behind glass at the back	missed	11.6 s
Hidden behind another	missed	missed
Tracks reported	5	4
Distinct people found	3 of 5	4 of 5
Mean dwell time	4.3 s	14.1 s

and close to a coin toss for the person at the left desk, with a confidence of 0.56. Age confidence stayed below 0.45 for everyone.

V. Discussion

A. Staged Transfer Learning

The probing result agrees with Kumar et al. [11]. A new head starts with random weights, and its first gradients are large and carry little information. If the backbone is trainable at that point those gradients flow into it. Training the head alone first removes the problem at the cost of a few short epochs.

The preferred weight decay of 0.3 is higher than the range of 0.01 to 0.1 often used with AdamW. PETA has 19,000 images and ViT-B/16 has 86 million parameters, so a need for stronger regularisation is plausible. The measured differences are small, and each rests on one run. Repeating the study with several random seeds is the first thing we would do with more compute.

B. Reading the Age Results

The oldest group is the easiest on every measure. Grey or white hair changes the colour histogram of the top region, and posture and clothing may add to the signal. This is our reading of the result. We did not test which features carry it.

The two youngest groups are confused with each other more than any other pair. A person of 29 and a person of 32 fall on opposite sides of a boundary and look alike at surveillance resolution. The confusion matrix supports this view of the errors as boundary effects. Four errors in five are off by one group, and errors of two or three groups make up 4.8% of the test set. For a store that wants a rough age profile of its visitors, an error of one group matters far less than an error of three. A measure that respects the order of the groups would credit the model for this, and mean accuracy does not.

The oldest group has only 8% of the data and still reaches a recall of 0.85. Balanced class weights push each SVM to find its positives, and this result suggests they are doing that job.

Leaving out the under 15 group reflects a wider problem. Surveillance datasets hold few children, both because few appear in the footage and because privacy rules limit how such data is collected. A store that needs to know how many children visit should not rely on IFAS for that.

C. Feature Settings and the Released Model

The released model and the selected settings differ in two ways, the number of regions and the number of LBP points. On validation data the difference in mean mA is 0.002, which is too small to separate the two on a single split. The screening study still earns its place. It shows that the model is not fragile. Across every setting we tried, average mA stays between 0.7155 and 0.7493, a spread of 3.4 points, and the two settings that cost most are too few body regions and too many colour bins.

D. Why Two Kinds of Model

One network that predicts gender and age together would be simpler to deploy. We kept the two apart for two reasons. The age model reproduces a published baseline, so its numbers can be set directly against those of Deng et al. [2]. Its features can also be inspected, since a histogram bin is a colour range in a named part of the body. The gender task has enough data to train a large network, and the accuracy of the transformer there is hard to match with hand-built features.

The runtime figures add a third argument. The age model costs less than a tenth of the gender network per crop on a processor. A store with no graphics card can run age on every crop and spend its budget on gender.

E. Limitations

Apart from the field check, the numbers in this paper come from PETA. The field check is one clip of 15 s labelled by eye, so accuracy in a real store is still unmeasured. Several other limits follow from the design.

- The tracker does not re-identify people by appearance. A person who leaves the frame and returns is counted twice, and two people who cross can swap tracks. The rule that rejoins lost tracks is a heuristic and can merge two people who stand in the same spot a few seconds apart.
- A person no detector finds is never counted. In the field check one person of five, hidden behind another, was missed for the whole clip.
- The tracking rules were tuned on a single recording. They need to be checked on labelled footage from other stores.
- Occupancy is the number of people in view, not the number in the store.
- Dwell time measures presence in the frame. It stands in for interest and does not measure it.
- The age SVMs were trained at the resolution of PETA images. Small or blurred crops receive labels with low confidence.
- Accuracy per track has not been measured. Pooling helps only when the errors on different frames are independent.
- The gender results are validation accuracy from short single runs. The gender model has no held-out test result yet.
- The runtime figures come from one processor and one recording.

VI. Conclusion

IFAS turns store footage into a record of who came in, when they came and how long they stayed. It does so with a mix of models chosen task by task: a small detector for speed, a fine-tuned transformer for gender and an SVM baseline for age. On PETA the gender model reaches 97.8% validation accuracy. The age model reaches a mean mA of 0.835 on the test split, and 95% of its predictions are within one age group of the truth. The sensitivity study shows that linear probing before fine-tuning has the largest effect on the gender model and that the number of body regions has the largest effect on the age model. The timings show that the gender network accounts for most of the cost and that capping the votes per person keeps that cost bounded. On a hard clip from outside PETA, rejoining lost tracks and checking weak detections with a second detector raised the number of people found from three of five to four of five and brought the reported dwell times in line with the footage.

Three pieces of work come next. The first is to label store footage and measure accuracy per track on it. The second is to add re-identification so that returning visitors are counted once. The third is to run the pipeline on a live camera feed, which we have not yet tested.

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